

2-Channel Time Interleaved ADC

Team A03 Aevulog

About Team

Name	Discord name	Affiliation (experience)	Role
Shreyas Angadi	Shreyas	Georgia Institute of Technology (Undergrad)	Digital Designer (Team Lead)
Andrés De La Torre	baggy_star	Georgia Institute of Technology (Undergrad)	Analog Designer
Logan Whitehead	Logan Whitehead	Georgia Institute of Technology (Undergrad)	Analog Designer
Lena Conde Araujo	Iena	Olin College of Engineering (Undergrad)	Analog Designer

Project Information

- Two 8-bit SAR ADCs are interleaved to achieve 50 MS/s throughput on GF180MCU, each channel sampling at 25 MS/s
- Each SAR ADC uses a double-tail comparator and a split-array CDAC, for lower power and faster operation
- Digital background calibration corrects offset and gain mismatch between channels without interrupting operation
- Review Date: 7/4/2026
- Track A
- Review Issue: [Issue #23](#)
- Review Repository: [chipathon-2026-ti-adc](#)

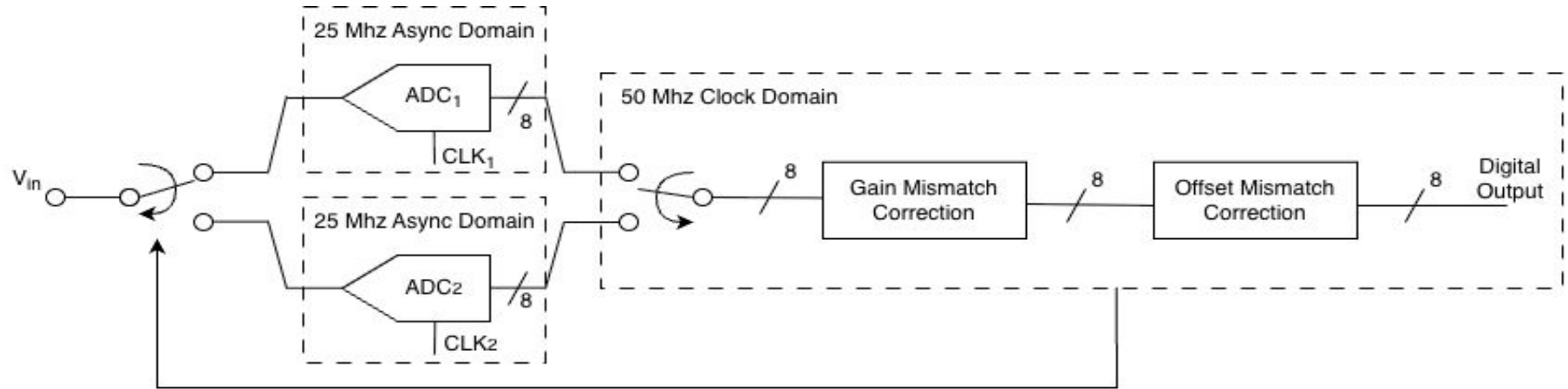
Design Objective

- The circuit enables faster sampling utilizing a SAR ADCs, doubling the sampling rate of an ADC
- It is intended to be used as a foundational ADC that can be used whenever a digital block needs to interact with analog voltages.

Target Specifications

Parameters	Specifications
Area	Block B
Sampling Rate	50 Ms / sec
Bits	8 bits
ENOB / SFDR	7.2 bits / 60 dBc
Input Range	0.0V - 3.3V

System Overview



Track and Hold Circuits Utilize same 25 Mhz
Clock Derived from 50 Mhz Clock Domain

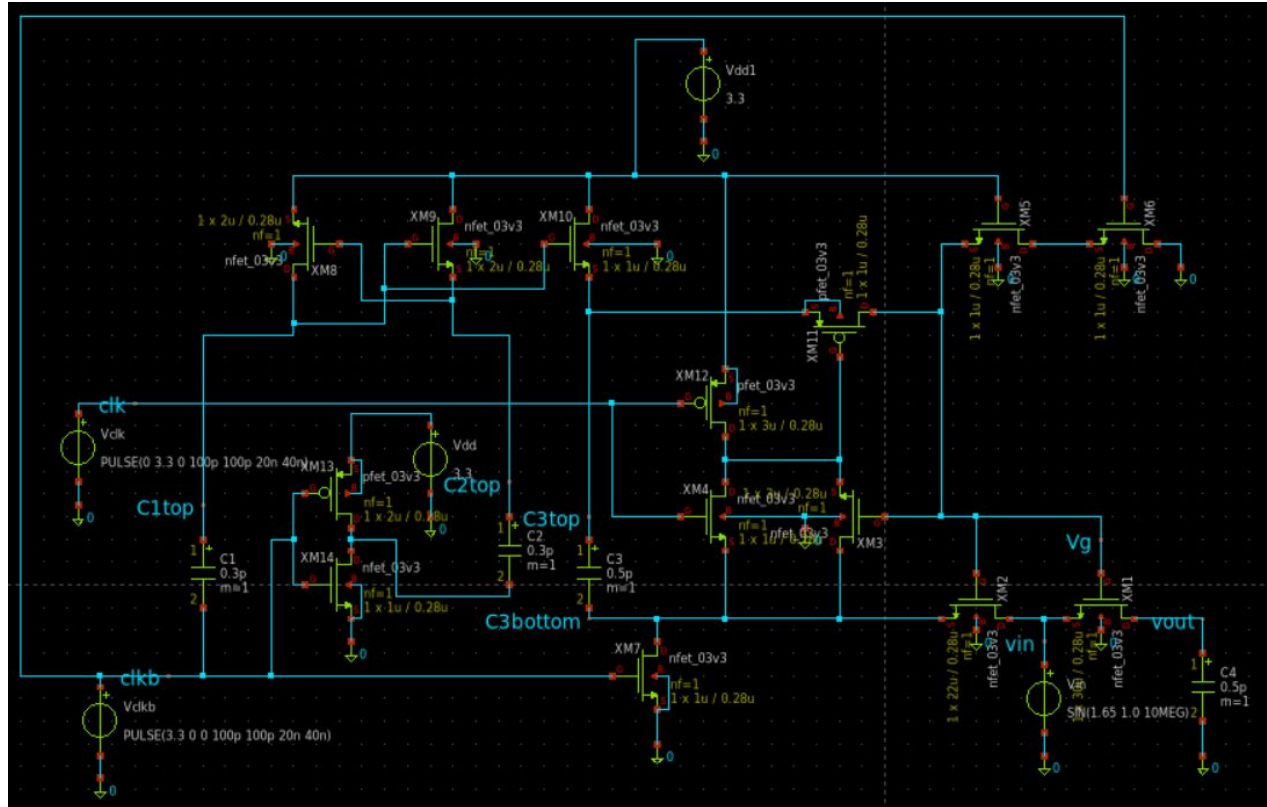
Purpose:

Hold input voltage steady while ADC makes conversion.

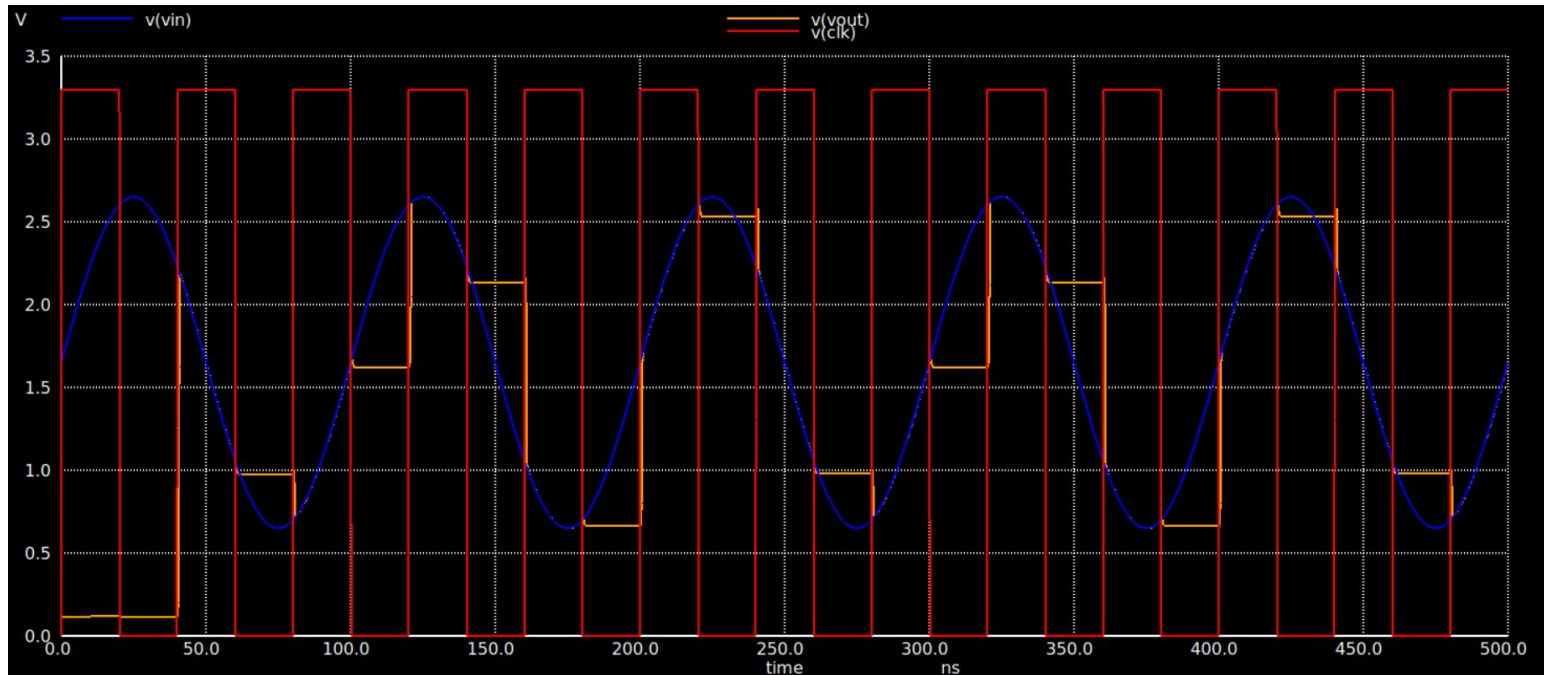
Inputs: VIN, CLK

Outputs: VOUT

Track and Hold Circuit



- The bootstrapped switch first precharges a capacitor to V_{dd} while the sampling nmos is off.
- During track, that capacitor is floated and stacked on top of the input signal, driving the sampling gate to approximately: $V_g = V_{in} + V_{dd}$



Double-Tailed Comparator

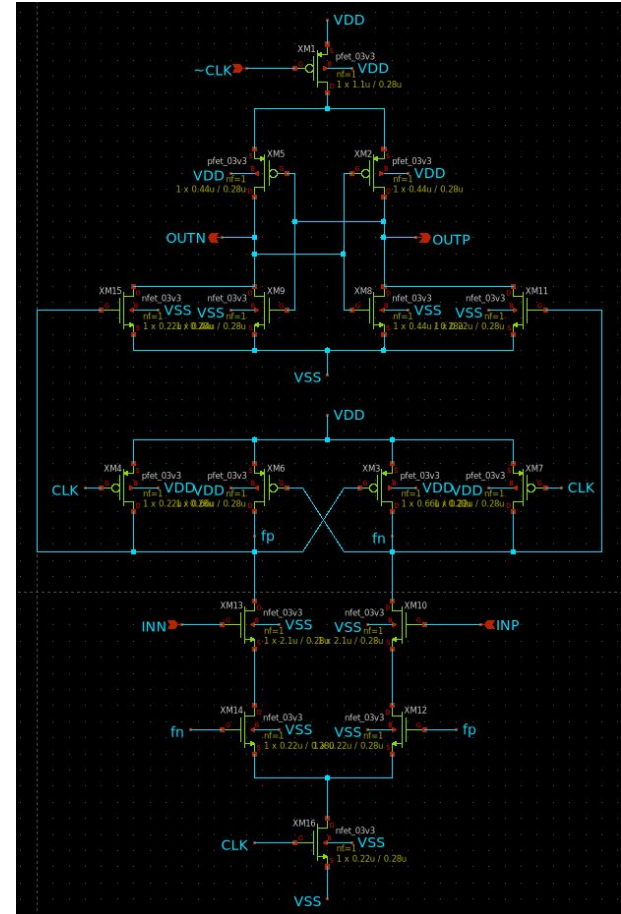
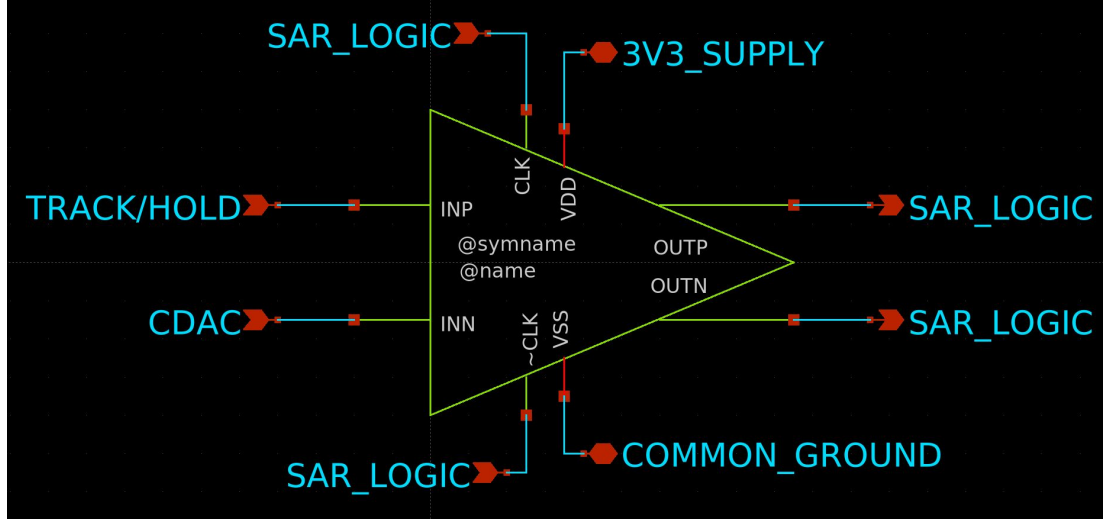
Purpose:

Compare the analog input from T/H to each consecutive binary-searched voltage

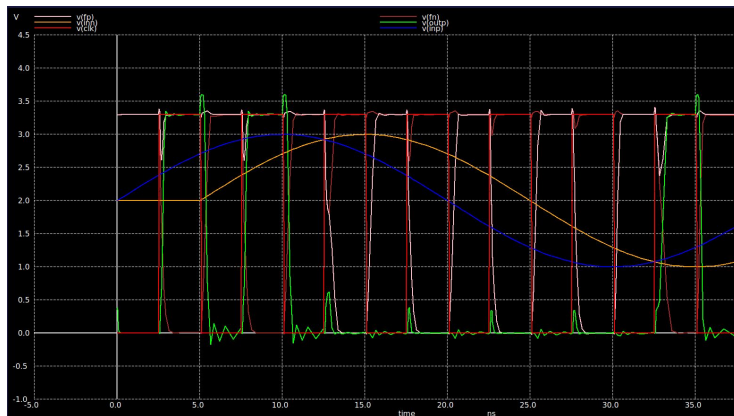
Inputs: compare-clk (200MS/s), INP & INN (0V-3.3V)

Outputs: OUTP, OUTN (0V-3.3V)

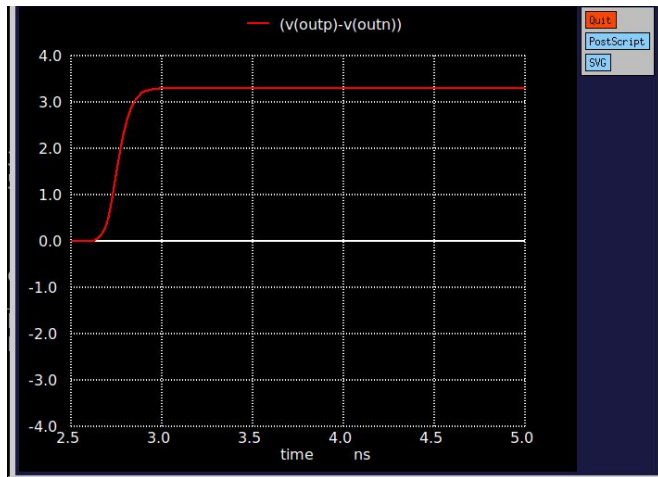
Interfaces with other blocks:



General Operation



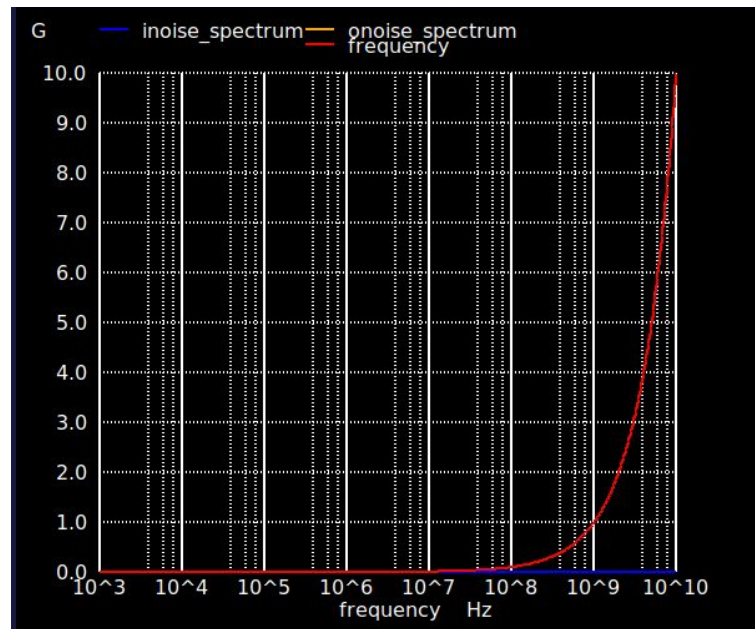
Metastability



Tau_reneg:
1.2 ns
2 time
constants of
growth

Design Decisions

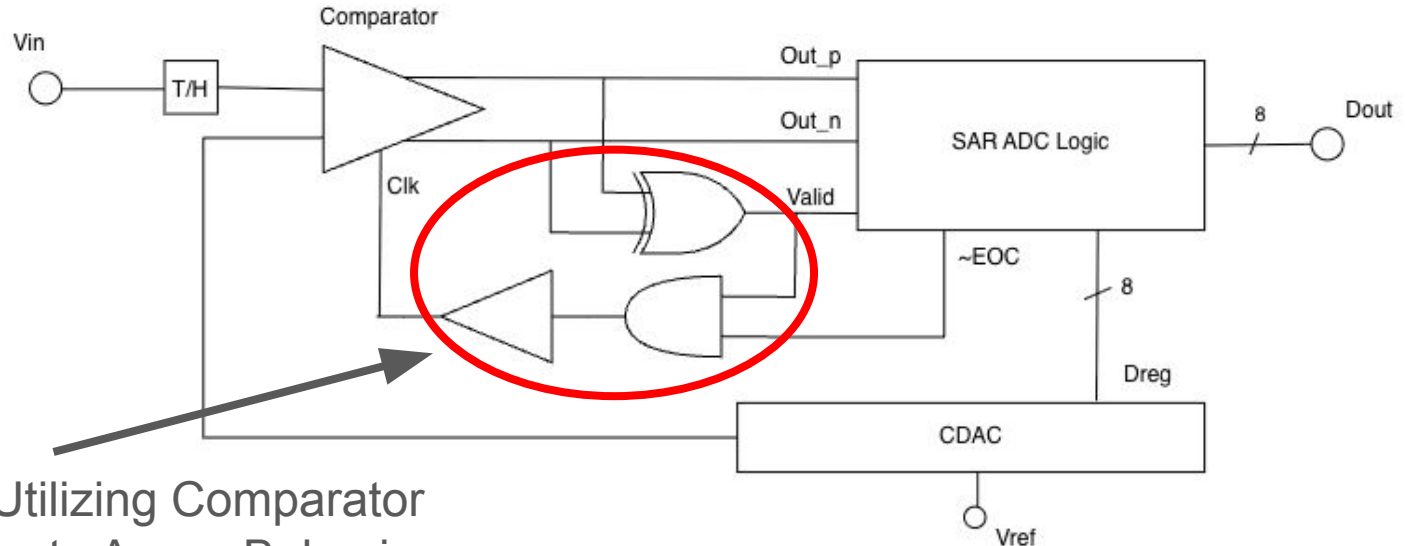
Gain of Noise across frequencies



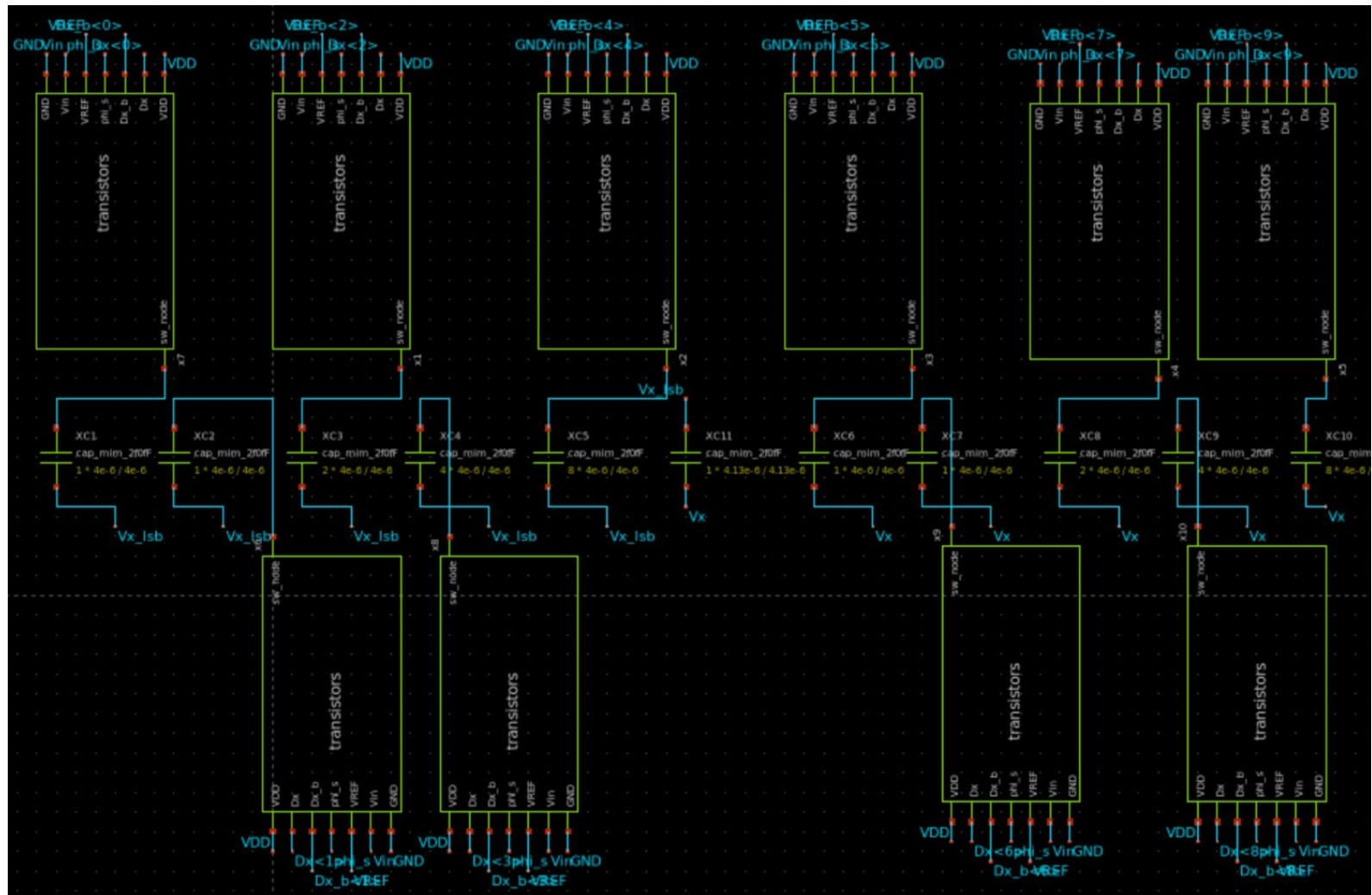
SAR ADC Logic

Purpose: Execute binary search logic of the SAR ADC

Inputs: Comparator Differential Outputs, **Outputs:** Dout, Dreg, Valid, EOC



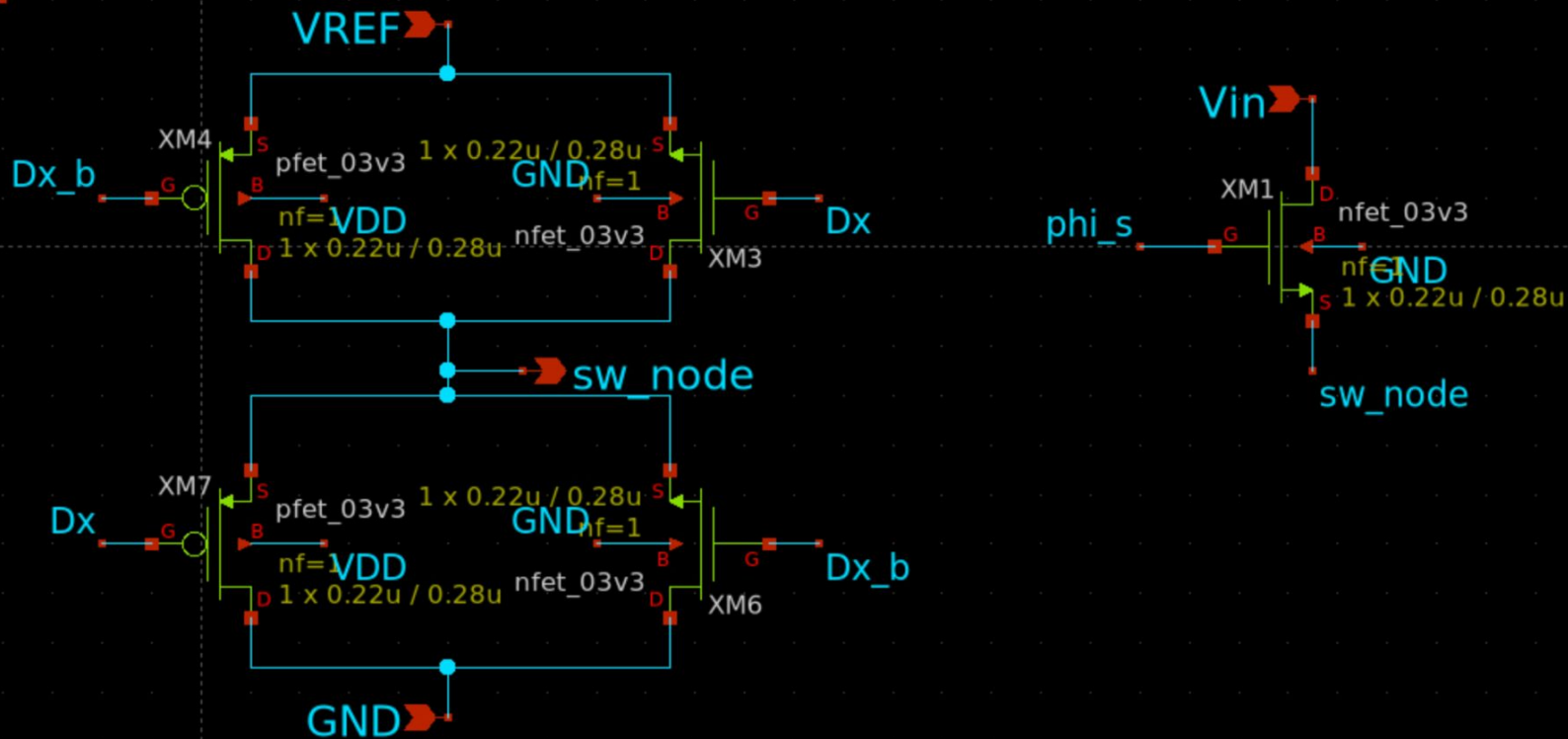
Custom Clock Utilizing Comparator Output to Generate Async Behavior



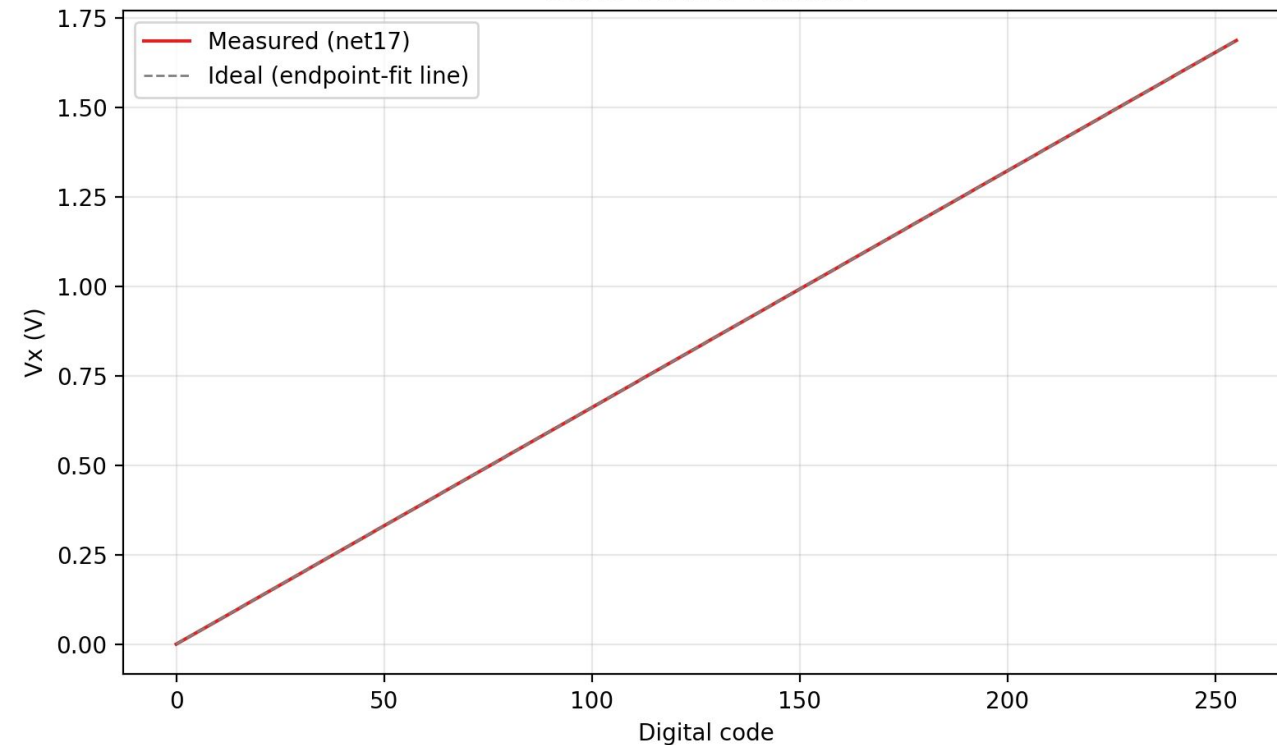
Dx → VDD →

Dx_b →

phi_s →



CDAC Transfer Function

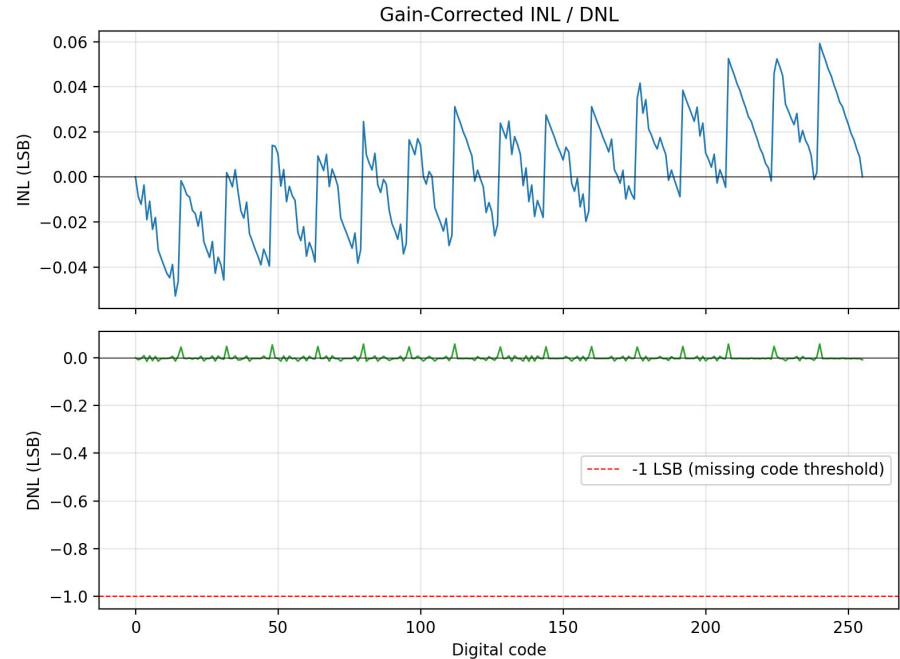


- Monotonic ramp across all 256 codes, full-scale sweep
- Zero missing codes (DNL never crosses -1 LSB)

INL/DNL (gain-corrected)

- DNL: +0.057 / -0.015 LSB
- INL: +0.059 / -0.053 LSB
- Gain error: -5.88% (systematic, calibratable)

Sub-0.06 LSB INL/DNL show the array is highly linear; the -5.88% gain error is a fixed, correctable offset



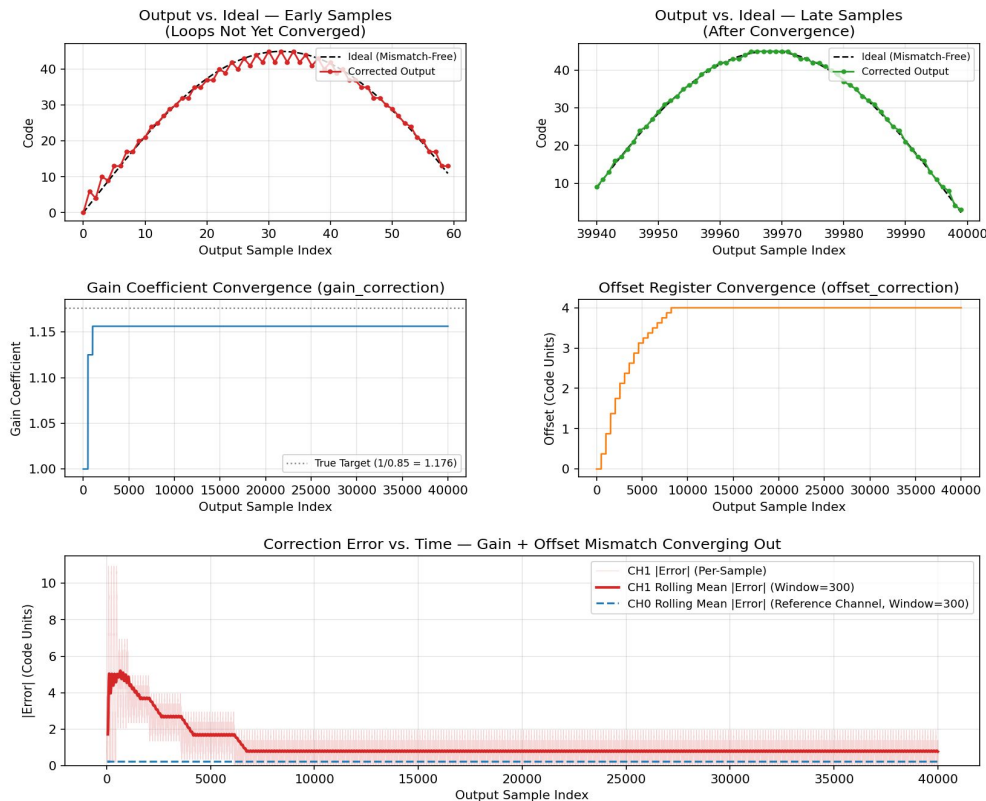
Gain Mismatch Digital Correction

- Dynamically corrects inter-channel gain mismatch in a time-interleaved architecture by treating CH0 as the reference and scaling CH1 to match its signal envelope.
- Accumulates the absolute values of both ADC channels over a configurable sample window to extract the average signal magnitude and calculate the instantaneous gain error.
- Adjusts a fixed-point gain coefficient continuously using an LMS-style feedback loop, utilizing a computationally cheap, right-shifted step size (μ) instead of complex division.
- Features an activity-monitoring threshold to freeze the calibration loop during DC or low-signal periods, preventing coefficient divergence, alongside a 2-stage pipelined datapath with saturation logic to prevent overflow.

Offset Mismatch Digital Correction

- Compensates for DC offset mismatch between time-interleaved channels by dynamically aligning the baseline of CH1 to the CH0 reference.
- Averages incoming signed samples over a configurable, power-of-two window to filter out AC signal components and isolate the instantaneous DC error.
- Updates the offset correction value using a computationally light, shift-based step size (μ) into a high-precision register, preserving fractional bits to ensure stable, fine-grained convergence.
- Subtracts the rounded, integer-converted offset from CH1 in a pipelined stage, utilizing built-in saturation logic to protect the final output against numerical overflow or wrapping.

correction_top (8-Bit Datapath): Combined Gain + Offset LMS Calibration Converging on a 15% Gain Mismatch + 4-Code Offset Mismatch Injected on ADC1



Digital Correction

- Parameterizable RTL for both gain correction and offset correction finished
- Final parameters need to be determined via simulations
- Sanity Testbenches show promising results of being able to remove injected errors
 - More thorough verification needs to be completed
- Netlist and sanity testbenches are viewable on our [Github Repository](#)

Design Checklist

Digital Tasks	Status
System Architecture and Design Specs	Completed
RTL Design	Completed
RTL Verification	In Progress
Floor Plan + Layout	In Progress

Analog Tasks	Status
Schematic Design	Completed
Spice Simulations	Completed
Layout Design	In Progress
Parasitic Simulations	In Progress
Mixed-Signal Integration	Not started

Design Assumptions

- Supply Voltages: 3.3V
- Temperature Range: 25°C - 125°C
- Input Range 0.0V - 3.3V
- External Circuitry: Input Voltage to be measured

Questions

1. Should we change comparator topology to be able to input negative voltages?
2. Do you have any recommendations for how to thoroughly verify all of the digital blocks?
3. After simulating and verifying every individual block, what is the best way to integrate them all together?
4. What should we as a team focus on to ensure that we get to Tapeout / what are the most pressing challenges we aren't considering?

Resources

- J. Elbornsson, F. Gustafsson and J. -E. Eklund, "Blind adaptive equalization of mismatch errors in a time-interleaved A/D converter system," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 51, no. 1, pp. 151-158, Jan. 2004
- Daihong Fu, K. C. Dyer, S. H. Lewis and P. J. Hurst, "A digital background calibration technique for time-interleaved analog-to-digital converters," in *IEEE Journal of Solid-State Circuits*, vol. 33, no. 12, pp. 1904-1911, Dec. 1998
- Splitarray origin: B. Ginsburg & A. Chandrakasan, "500-MS/s 5-bit ADC in 65-nm CMOS," *JSSC* 2006
- Hariprasath et al., "Merged capacitor switching based SAR ADC," *Electronics Letters* 2010
- "Capacitor mismatch effects in SC circuits," *ISCAS* 1990
- S. Babayan-Mashhadi and R. Lotfi, "Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator," in *IEEE Transactions on Very Large Scale Integration (VLSI)*

